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JEE Main Physics Practice Sheet — DPP-028

Topic: Equilibrium of Concurrent Forces & Lami's Theorem

Time Allowed: 60 Minutes — **Max Marks:** 80 — **Marking Scheme:** +4, -1

- Q1.** Three concurrent coplanar forces $\vec{F}_1, \vec{F}_2, \vec{F}_3$ keep a particle in static equilibrium. If the angle between $\vec{F}_1$ and $\vec{F}_2$ is 90° , and the angle between $\vec{F}_2$ and $\vec{F}_3$ is 120° , the ratio of the magnitudes $|\vec{F}_1| : |\vec{F}_2| : |\vec{F}_3|$ is:
- (A) $1 : \sqrt{3} : 2$
 (B) $\sqrt{3} : 1 : 2$
 (C) $1 : 1 : \sqrt{2}$
 (D) $\sqrt{3} : 2 : 1$
- Q2.** A body of mass $m = 10$ kg is suspended by two light strings of lengths 3 m and 4 m attached to two points on a horizontal ceiling 5 m apart. The tension in the 3 m string is ($g = 10$ m/s²):
- (A) 60 N
 (B) 80 N
 (C) 100 N
 (D) 75 N
- Q3.** A block of weight $W = 100$ N is suspended by a string from a rigid support. The string is pulled horizontally by a force F such that it makes an angle of 30° with the vertical. The magnitude of the horizontal force F and the tension T in the string are:
- (A) $F = \frac{100}{\sqrt{3}}$ N, $T = \frac{200}{\sqrt{3}}$ N
 (B) $F = 100\sqrt{3}$ N, $T = 200$ N
 (C) $F = 50$ N, $T = 100$ N
 (D) $F = \frac{50}{\sqrt{3}}$ N, $T = \frac{100}{\sqrt{3}}$ N
- Q4.** A smooth uniform sphere of weight W and radius R rests against a smooth vertical wall and is held by a string of length $L = R$ attached to the top of the sphere and to a point on the wall. The normal reaction exerted by the wall on the sphere is:
- (A) $\frac{W}{\sqrt{3}}$
 (B) $\frac{W}{2}$
 (C) $W\sqrt{3}$
 (D) $\frac{2W}{\sqrt{3}}$
- Q5.** Three concurrent forces $\vec{F}_1 = (2\hat{i} + 3\hat{j})$ N, $\vec{F}_2 = (-4\hat{i} + \hat{j})$ N, and $\vec{F}_3$ act on a particle. If the particle is in equilibrium, the force $\vec{F}_3$ must be:
- (A) $(2\hat{i} - 4\hat{j})$ N
 (B) $(-2\hat{i} + 4\hat{j})$ N
 (C) $(6\hat{i} - 2\hat{j})$ N
 (D) $(-6\hat{i} + 2\hat{j})$ N
- Q6.** A heavy particle of mass m is suspended by a string of length l . A horizontal force F is applied to the particle such that it is held in equilibrium with the string deflected at an angle θ to the vertical. The tension in the string is:
- (A) $mg \cos \theta$
 (B) $\frac{mg}{\cos \theta}$
 (C) $mg \sin \theta$
 (D) $\sqrt{(mg)^2 - F^2}$
- Q7.** Which of the following sets of three concurrent coplanar forces CANNOT produce equilibrium on a body?
- (A) 3 N, 4 N, 5 N
 (B) 2 N, 7 N, 10 N
 (C) 5 N, 5 N, 5 N
 (D) 10 N, 15 N, 20 N
- Q8.** A uniform smooth sphere of weight W is supported between two smooth inclined planes making angles 30° and 60° with the horizontal. The normal reactions N_1 and N_2 exerted by the 30° and 60° planes respectively are:
- (A) $N_1 = \frac{\sqrt{3}}{2}W$, $N_2 = \frac{1}{2}W$
 (B) $N_1 = \frac{1}{2}W$, $N_2 = \frac{\sqrt{3}}{2}W$
 (C) $N_1 = W$, $N_2 = W$
 (D) $N_1 = \frac{W}{\sqrt{3}}$, $N_2 = W\sqrt{3}$
- Q9.** A weight $W = 60$ N is suspended by two strings OA and OB connected at point O . OA makes an angle of 45° with the horizontal and OB makes an angle of 45° with the vertical. The tension in each string is:
- (A) $30\sqrt{2}$ N
 (B) $60\sqrt{2}$ N
 (C) 30 N
 (D) 60 N
- Q10.** Three forces acting on a particle are represented in magnitude and direction by the three sides of a triangle taken in order. The resultant of these forces is:
- (A) Infinity
 (B) Proportional to the perimeter of the triangle
 (C) Zero
 (D) Double the area of the triangle

- Q11.** A smooth ring P of mass m is threaded on a smooth circular wire of radius R fixed in a vertical plane. A light elastic cord of natural length R has one end fixed to the highest point A of the wire and the other to P . If the ring is in equilibrium at an angle θ from the vertical at A , the normal reaction of the wire on the ring:
- (A) Points radially inward
(B) Points radially outward
(C) Is zero
(D) Equals mg
- Q12.** Two light strings AC and BC are tied to a small block of mass $m = 5$ kg at C . String AC makes 60° with the horizontal ceiling and string BC makes 30° with the vertical wall. If the system is in equilibrium, the ratio of tension T_{AC} to T_{BC} is:
- (A) 1 : 1
(B) $\sqrt{3} : 1$
(C) $1 : \sqrt{3}$
(D) 2 : 1
- Q13.** A cylinder of weight $W = 200$ N rests in a V-shaped groove whose smooth sides make angles of 30° and 60° with the vertical. The reaction force exerted on the cylinder by the steeper side (making 30° with the vertical) is:
- (A) 100 N
(B) $100\sqrt{3}$ N
(C) 200 N
(D) $\frac{200}{\sqrt{3}}$ N
- Q14.** A picture frame of mass m is hung from a nail on a vertical wall by a symmetric cord of total length L attached to its two top corners separated by distance d ($d < L$). The tension in the cord is:
- (A) $\frac{mgL}{2\sqrt{L^2 - d^2}}$
(B) $\frac{mgd}{2\sqrt{L^2 - d^2}}$
(C) $\frac{mgL}{\sqrt{L^2 - d^2}}$
(D) $\frac{mg}{2}$
- Q15.** In Lami's theorem, if three coplanar forces $\vec{P}, \vec{Q}, \vec{R}$ maintain equilibrium and the angles opposite to them are α, β, γ respectively, which condition is mandatory for the formula $\frac{P}{\sin \alpha} = \frac{Q}{\sin \beta} = \frac{R}{\sin \gamma}$ to be valid?
- (A) The three forces must be collinear
(B) The forces must be concurrent, non-collinear, and coplanar
(C) The forces must be mutually perpendicular
(D) The angles α, β, γ must each be obtuse
- Q16.** A smooth sphere of mass M is placed between two smooth parallel vertical walls separated by a distance slightly greater than its diameter. A horizontal force F pushes the sphere against one wall. The net vertical normal reaction from the smooth floor on the sphere is:
- (A) $Mg + F$
(B) Mg
(C) $\sqrt{(Mg)^2 + F^2}$
(D) Zero
- Q17.** A block of mass m rests on a smooth inclined plane of inclination θ . It is held in equilibrium by a force F acting horizontally. The normal reaction exerted by the incline on the block is:
- (A) $mg \cos \theta$
(B) $\frac{mg}{\cos \theta}$
(C) $mg \sin \theta$
(D) $mg \tan \theta$
- Q18.** A heavy bead of mass m is threaded on a smooth vertical circular ring of radius R . A light inextensible string attached to the bead passes over a small smooth peg at the top of the ring and supports a hanging mass M . For equilibrium of the bead at an angle θ subtended at the top peg with the vertical, the ratio M/m must be:
- (A) $2 \cos \theta$
(B) $2 \sin \theta$
(C) $\cos \theta$
(D) 1
- Q19.** Three concurrent forces F_1, F_2, F_3 satisfy $\vec{F}_1 + \vec{F}_2 + \vec{F}_3 = \vec{0}$. If $|F_1| = 5$ N and $|F_2| = 12$ N, and the angle between $\vec{F}_1$ and $\vec{F}_2$ is 90° , then the magnitude of $\vec{F}_3$ is:
- (A) 7 N
(B) 13 N
(C) 17 N
(D) 60 N
- Q20.** A small metallic sphere of mass $m = 2$ kg carrying charge q is suspended by an insulating thread of length l . An external horizontal uniform electric field produces a constant electrostatic force $F_e = 20$ N on the sphere. At equilibrium, the angle made by the thread with the vertical is ($g = 10$ m/s²):
- (A) 30°
(B) 45°
(C) 60°
(D) 90°

SMAPhysics.com — Answer Key & Detailed Solutions (DPP-028)

Equilibrium of Concurrent Forces & Lami's Theorem

Q.No.	Ans	Q.No.	Ans	Q.No.	Ans	Q.No.	Ans
1	B	6	B	11	A	16	B
2	B	7	B	12	B	17	B
3	A	8	A	13	A	18	A
4	A	9	A	14	A	19	B
5	A	10	C	15	B	20	B

Detailed Step-by-Step Solutions

Sol 1. Option (B)

The angle between $\vec{F}_1$ and $\vec{F}_2$ is $\theta_{12} = 90^\circ$.

Angle between $\vec{F}_2$ and $\vec{F}_3$ is $\theta_{23} = 120^\circ$.

The third angle between $\vec{F}_3$ and $\vec{F}_1$ is $\theta_{31} = 360^\circ - (90^\circ + 120^\circ) = 150^\circ$.

By Lami's theorem:

$$\frac{F_1}{\sin \theta_{23}} = \frac{F_2}{\sin \theta_{31}} = \frac{F_3}{\sin \theta_{12}}$$

$$\frac{F_1}{\sin 120^\circ} = \frac{F_2}{\sin 150^\circ} = \frac{F_3}{\sin 90^\circ}$$

Since $\sin 120^\circ = \frac{\sqrt{3}}{2}$, $\sin 150^\circ = \frac{1}{2}$, and $\sin 90^\circ = 1$:

$$F_1 : F_2 : F_3 = \frac{\sqrt{3}}{2} : \frac{1}{2} : 1 = \sqrt{3} : 1 : 2$$

Sol 2. Option (B)

The lengths of the strings and ceiling distance form a right-angled triangle since $3^2 + 4^2 = 5^2$.

Let θ_1 be the angle the 3 m string makes with the ceiling: $\cos \theta_1 = 3/5 = 0.6 \implies \sin \theta_1 = 4/5 = 0.8$.

Angle made by the 3 m string with the vertical is α , where $\cos \alpha = \sin \theta_1 = 4/5 = 0.8$.

Let the tension in the 3 m string be T_1 and in the 4 m string be T_2 .

Since the two strings are perpendicular to each other (90° between them): Resolving vertically:

$$T_1 \cos \alpha + T_2 \sin \alpha = mg = 10 \times 10 = 100 \text{ N}$$

Resolving along the 3 m string:

$$T_1 = mg \cos \alpha = 100 \times \frac{4}{5} = 80 \text{ N}$$

Sol 3. Option (A)

The three concurrent forces acting on the knot are: 1. Downward weight $W = 100 \text{ N}$ 2. Horizontal force F 3. Tension T at angle 30° to vertical.

Resolving vertically:

$$T \cos 30^\circ = W \implies T \left(\frac{\sqrt{3}}{2} \right) = 100 \implies T = \frac{200}{\sqrt{3}} \text{ N}$$

Resolving horizontally:

$$F = T \sin 30^\circ = \left(\frac{200}{\sqrt{3}} \right) \left(\frac{1}{2} \right) = \frac{100}{\sqrt{3}} \text{ N}$$

Sol 4. Option (A)

The sphere has radius R . The distance from sphere center O to wall contact C is R .

The string is attached to the sphere's top and extends length $L = R$ to the wall.

The line from string wall anchor A through sphere center O has length $AO = R + L = 2R$.

In $\triangle AOC$, $\sin \theta = \frac{R}{2R} = \frac{1}{2} \implies \theta = 30^\circ$, where θ is the angle the string line makes with the vertical wall.

At center O , three concurrent forces act: Normal reaction N (horizontal), Weight W (vertically down), Tension T (at 30° to vertical).

$$T \cos 30^\circ = W \implies T = \frac{W}{\cos 30^\circ} = \frac{2W}{\sqrt{3}}$$

$$N = T \sin 30^\circ = \left(\frac{2W}{\sqrt{3}} \right) \left(\frac{1}{2} \right) = \frac{W}{\sqrt{3}}$$

Sol 5. Option (A)

For static equilibrium of concurrent forces:

$$\vec{F}_1 + \vec{F}_2 + \vec{F}_3 = \vec{0}$$

$$(2\hat{i} + 3\hat{j}) + (-4\hat{i} + \hat{j}) + \vec{F}_3 = \vec{0}$$

$$(-2\hat{i} + 4\hat{j}) + \vec{F}_3 = \vec{0} \implies \vec{F}_3 = (2\hat{i} - 4\hat{j}) \text{ N}$$

Sol 6. Option (B)

Resolving forces at the particle along the vertical:

$$T \cos \theta = mg \implies T = \frac{mg}{\cos \theta}$$

(Along horizontal: $T \sin \theta = F \implies F = mg \tan \theta$).

Sol 7. Option (B)

For three concurrent forces to be in equilibrium, the sum of any two magnitudes must be greater than or equal to the third magnitude ($F_1 + F_2 \geq F_3$ for the largest F_3 , triangle inequality of vectors).

For option (B): $2 + 7 = 9 < 10$. The forces cannot form a closed triangle, so equilibrium is impossible.

Sol 8. Option (A)

The normal reaction N_1 from the 30° plane acts at an angle 30° to the vertical.

The normal reaction N_2 from the 60° plane acts at an angle 60° to the vertical.

The angle between N_1 and N_2 is 90° .

Resolving forces along the direction of N_1 :

$$N_1 = W \cos 30^\circ = W \frac{\sqrt{3}}{2}$$

Along direction of N_2 :

$$N_2 = W \cos 60^\circ = \frac{1}{2}W$$

Sol 9. Option (A)

Both strings make 45° with the vertical.

By symmetry, tensions are equal: $T_1 = T_2 = T$.

Vertical equilibrium:

$$2T \cos 45^\circ = W = 60 \text{ N}$$

$$2T \left(\frac{1}{\sqrt{2}} \right) = 60 \implies \sqrt{2}T = 60 \implies T = \frac{60}{\sqrt{2}} = 30\sqrt{2} \text{ N}$$

Sol 10. Option (C)

If three vectors $\vec{A}, \vec{B}, \vec{C}$ form the three sides of a triangle in order, their vector sum is $\vec{A} + \vec{B} + \vec{C} = \vec{0}$ by the polygon law of addition. Thus, the resultant is zero.

Sol 11. Option (A)

Let O be the center of the circular wire. The triangle formed by O, A, P is isosceles with $OA = OP = R$.

The chord length AP is the tension line. The angle of the chord with the radial line OP is θ .

By symmetry and balance of forces along tangent and radial directions, the normal reaction from the wire acts purely along the radius towards the center (radially inward).

Sol 12. Option (B)

AC makes 60° with the horizontal $\implies 30^\circ$ with the vertical.

BC makes 30° with the vertical wall $\implies 60^\circ$ with the horizontal (30° with vertical in opposite sense).

Let horizontal angle of AC be 30° and BC be 60° : Horizontal equilibrium:

$$T_{AC} \cos 60^\circ = T_{BC} \sin 30^\circ \text{ (depending on orientation)}$$

Using Lami's theorem:

$$\frac{T_{AC}}{\sin 120^\circ} = \frac{T_{BC}}{\sin 150^\circ} \implies \frac{T_{AC}}{T_{BC}} = \frac{\sin 120^\circ}{\sin 150^\circ} = \frac{\sqrt{3}/2}{1/2} = \sqrt{3}$$

Sol 13. Option (A)

The side making 30° with the vertical exerts a normal reaction N_1 inclined at 60° to the vertical (30° to horizontal).

The other side makes 60° with the vertical, exerting normal reaction N_2 inclined at 30° to the vertical.

Since the two reaction forces are mutually perpendicular ($30^\circ + 60^\circ = 90^\circ$):

$$N_1 = W \cos 60^\circ = 200 \times \frac{1}{2} = 100 \text{ N}$$

Sol 14. Option (A)

The cord forms an isosceles triangle with base d and two equal sides $L/2$.

Let θ be the angle each half-cord makes with the vertical:

$$\sin \theta = \frac{d/2}{L/2} = \frac{d}{L} \implies \cos \theta = \sqrt{1 - \left(\frac{d}{L} \right)^2} = \frac{\sqrt{L^2 - d^2}}{L}$$

Vertical equilibrium:

$$2T \cos \theta = mg \implies T = \frac{mg}{2 \cos \theta} = \frac{mgL}{2\sqrt{L^2 - d^2}}$$

Sol 15. Option (B)

Lami's theorem states that if three coplanar, concurrent, and non-collinear forces keep a body in equilibrium, the magnitude of each force is proportional to the sine of the angle between the other two forces.

Sol 16. Option (B)

The walls are vertical and frictionless (smooth), so they can only exert horizontal normal reactions.

The floor is smooth and horizontal, exerting a vertical normal reaction N_{floor} .

Vertical equilibrium of the sphere:

$$\sum F_y = 0 \implies N_{\text{floor}} - Mg = 0 \implies N_{\text{floor}} = Mg$$

Sol 17. Option (B)

Resolving forces along the inclined plane:

$$mg \sin \theta = F \cos \theta \implies F = mg \tan \theta$$

Resolving forces perpendicular to the inclined plane:

$$N = mg \cos \theta + F \sin \theta = mg \cos \theta + (mg \tan \theta) \sin \theta$$

$$N = mg \left(\cos \theta + \frac{\sin^2 \theta}{\cos \theta} \right) = mg \left(\frac{\cos^2 \theta + \sin^2 \theta}{\cos \theta} \right) = \frac{mg}{\cos \theta}$$

Sol 18. Option (A)

Tension in the string is $T = Mg$.

For the bead on the ring, the angle subtended by the bead with the vertical at the peg is θ .

The triangle formed by peg, center, and bead is isosceles, so the angle made by the normal reaction (along radius) is related to θ .

Balancing forces along the tangential direction to the ring:

$$T = 2mg \cos \theta \implies Mg = 2mg \cos \theta \implies \frac{M}{m} = 2 \cos \theta$$

Sol 19. Option (B)

Since $\vec{F}_1 + \vec{F}_2 + \vec{F}_3 = \vec{0}$, we have $\vec{F}_3 = -(\vec{F}_1 + \vec{F}_2)$.

Magnitude:

$$|\vec{F}_3| = \sqrt{F_1^2 + F_2^2 + 2F_1F_2 \cos 90^\circ} = \sqrt{5^2 + 12^2} = \sqrt{25 + 144} = \sqrt{169} = 13$$

Sol 20. Option (B)

Three concurrent forces act on the sphere: 1. Weight $W = mg = 2 \times 10 = 20 \text{ N}$ (downward) 2. Electric force $F_e = 20 \text{ N}$ (horizontal) 3. Tension T in the string at angle θ to the vertical.

Equilibrium gives:

$$T \sin \theta = F_e = 20 \text{ N}, \quad T \cos \theta = mg = 20 \text{ N}$$

$$\tan \theta = \frac{F_e}{mg} = \frac{20}{20} = 1 \implies \theta = 45^\circ$$